For my CMPSC 132 Project, I chose the “Number Guessing Game” option. In the project, I created a single terminal game where the system generates a random number after importing the “random” library, and this is labeled as the target number. Once this number is created, the user must try to guess the exact number with a number of attempts. The user must enter a valid integer digit, or else a message saying “Your guess should be a valid digit!” will be displayed on the console. If the user guesses a number that has a lower value than the target number, a message saying “Too Low” is printed as a string to indicate the user must enter a greater number. On the other hand, if the user guesses a number that has a higher value than the target number, a message saying “Too High” is printed as a string to indicate the user must enter a smaller number. If the user is able to guess the target number correctly, then a message saying “Correct!” is displayed, along with a message that tells the user the number of attempts they made leading up to guessing the target number correctly. The correct_guess variable is now set to “True”. Additionally, a user’s “guess_history” is displayed at the end that gives more detail each individual attempt made and the specific guess made by the user before guessing the correct target number. The number of attempts the user has made is tracked with the initialization variable called “num_attempts” and it is accumulated by 1 each time the user guesses a number that does not match up with the target number. The correct_guess variable is initialized to “False” to indicate that once the game starts, the user has not guessed the correct number yet. A “guess_history” is kept tracked with an empty list. Say the user does not guess the target number correctly and runs out of attempts. Then, the game ends and the user loses.

As for the extra features, I implemented an additional selection of three game modes for the game: easy, medium, and hard. In Easy Mode, the target number ranges from 1 to 50, the user has unlimited attempts to guess the number, and a hint is given to the user that says whether the target number is odd or even. In Medium Mode, the target number ranges from 1 to 100, the user has 20 attempts to guess the number, and a hint is given to the user that says whether the target number is odd or even. In Hard Mode, the target number ranges from 1 to 200, the user has 10 attempts to guess the number, and no hint is given to the user that says whether the target number is odd or even. Lastly, I implemented a “Play Again” feature in the main function to either reset or exit the game whether the user ended up guessing the target number correctly or not. Based on if the user inputs “yes” or “no”, the game will start a new round or terminate!